

EMMANUEL BOLARINWA

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EDUCATIONAL QUALIFICATION

2017 - 2023	B.Sc. Computer Engineering <i>Obafemi Awolowo University, Ife, Nigeria</i> (Ranked #3 in Nigeria) Class of Degree: Second Class Honors (Upper Division)
2010 - 2016	WASSCE, Molaks High School

RESEARCH EXPERIENCE

2024 - Present	AI Researcher, LyngualLabs - Participate in the collection of 100 hours of high-quality code-switched audio and transcribe datasets across 10+ diverse domains - Research, design, and develop AI models to tackle code-switching, low-resource languages, and language diversity through experiments.
2023 - 2024	Undergraduate Research, Pervasive and Mobile Computing Research Group (PMCRG) <u>Thesis Title</u> : Design and implementation of a low-cost electronic nose system <u>Supervisor</u> : Dr S Aina <u>Contribution</u> : Synthesized state-of-the-art research to guide the design and implementation of a low-cost electronic nose, engineering a complete machine learning pipeline with Principal Component Analysis (PCA) and Gradient Boosting for gas classification.
2021	Research Intern, Microelectronics Research Group - Conducted a comprehensive review of machine learning research on gas sensors and optimized algorithms for a gas sensor dataset, achieving improved accuracy, speed, and performance. - Deployed a real-time facemask detection model on the SIPEED Maixduino device.

TEACHING EXPERIENCE

2025 - Present	Machine Learning Tutor, Lyngual Labs Academy - Delivered engaging live sessions and hands-on labs in a beginner-friendly Machine Learning program, covering foundational Python, data science principles, and advanced topics in feature engineering and model interpretation. - Mentored students through the complete machine learning lifecycle, from Python basics and data wrangling to advanced model tuning and experiment tracking with MLflow, culminating in the development of portfolio-ready projects.
2024	Invited Lecturer, International Center for Hydropower (ICH) Workshop - Designed and delivered a technical lecture on modeling energy system uncertainties using Monte Carlo simulations to a diverse cohort of professionals and academics from 8+ African nations, becoming the youngest invited lecturer in the organization's history. - Mentored participants through practical, hands-on sessions, guiding them to apply advanced tools (PyPSA, Snakemake, GIS) to build transparent and replicable energy models, directly supporting ICH's mission to advance renewable energy capacity in the Global South.
2023 - 2024	Data Analyst Instructor, IJAN Africa - Architected and instructed a 4-month, project-based Data Analysis curriculum for a cohort of 20+ students, focusing on transforming raw data into actionable business insights. - Mentored students in practical data analysis techniques, guiding them to develop job-ready skills and a portfolio to secure roles on platforms like LinkedIn and Upwork, as emphasized by IJAN Africa's career-focused mission.
2023 - 2024	Teaching Assistant, Obafemi Awolowo University Instructed over 100 students in Python programming using a hybrid model, delivering engaging virtual lectures and hands-on in-person guidance to build their practical coding abilities.

PROFESSIONAL EXPERIENCE

2023 - Present	Energy System Modeler, Open Energy Transition, Bayreuth, Germany. - Developed and enhanced the PyPSA-Earth framework by engineering a Concentrated Solar Power (CSP) modeling functionality with thermal storage and implementing a Monte Carlo simulation module to quantify system uncertainties. - Advanced the transition to open-source data by conducting a comparative GIS analysis of ENTSO-E and OpenStreetMap (OSM) to improve model transparency. Contributed to developing a fully sector-coupled U.S. energy system model.
2022	Data Engineering/MLOps Intern, Hamoye, Lagos State, Nigeria. Developed an end-to-end machine learning project for quantitative data analysis that achieved a top 5 ranking for its design and implementation.
2021	Backend Intern, Zuri, Lagos State, Nigeria. Built and maintained a Django-based blog application, focusing on server-side logic, user authentication, and database management.
2020 - 2023	Web Developer, XLevelRetail, Lagos State, Nigeria.. Led the end-to-end development of four e-commerce websites from scratch using WordPress, integrating custom features and third-party plugins like WooCommerce to deliver high-performance online stores.

ACHIEVEMENTS/AWARD

2025	Paper Award, Deep Learning Indaba 2025 An award recognizing the top 1% of submitted papers for outstanding contribution and innovation in the field of African machine learning.
2024	Recognition for Research Contribution, Xiong et al. (2024) Formally acknowledged for substantial early developments and validation work on the integration of OpenStreetMap (OSM) data into the PyPSA-Earth energy modeling framework.
2024	Invited Lecturer (Youngest in Workshop History), International Centre for Hydropower (ICH) Workshop Selected by the ICH to present on the Effect of modelling uncertainties in the Energy System using Monte-Carlo to an international audience of academics and industry professionals.

PUBLICATIONS & PRESENTATIONS

2025	Co-presented a technical talk demonstrating the application of open-source tools, PyPSA-Earth and OpenStreetMap, to architect transparent and replicable models for sustainable energy planning in Africa. [Open Source Conference Africa]
2025	Bolarinwa, E., Babatunde, O., Olufemi, V., Moshood, K., & Williams, O. (2025). Closing the Gap in Low-Resource ASR: Leveraging Multilingual Models for Code-Switched Yoruba-English Speech. (Paper will be published in PMLR)
2025	Babatunde, O. B., Olufemi, V. T., Bolarinwa, E. T., Moshood, K. Y., & Emezue, C. C. (2025). Beyond Monolingual Limits: Fine-Tuning Monolingual ASR for Yoruba-English Code-Switching. Seventh Workshop on Computational Approaches to Linguistic Code-Switching (CALCS 2025). Beyond Monolingual Limits: Fine-Tuning Monolingual ASR for Yoruba-English Code-Switching - ACL Anthology
2025	Babatunde, O. B., Olufemi, V. T., Bolarinwa, E. T., & Moshood, K. Y. (2025). Challenging Multimodal LLMs with African Standardized Exams: A Document VQA Evaluation. ACL 2025 Workshop AfricaNLP Submission. Challenging Multimodal LLMs with African Standardized Exams: A Document VQA Evaluation
2024	Aina, S., Gambo, I. P., Ogungbe, O. A., Bolarinwa, E., & Oluwaranti, A. I. (2024). A low-cost electronic nose device. International Journal of Information Research and Review, 11(5), 7774-7777. https://ijirr.com/sites/default/files/issues-pdf/4040.pdf

SKILLS AND INTERESTS

Technical Skills: Python, Machine Learning, Deep Learning with special interest in NLP, Large Language Models (LLMs), PyPSA, Osemosys, Grafana, Matplotlib, Seaborn, Looker, PowerBI, MLflow, Docker, Git, Localstack, PySpark, PostgreSQL, MySQL, BigQuery, Terraform, dbt, Prefect

Tools: Jupyter Notebook, VSCode, Git, GitHub, Google Cloud, AWS, Azure

Academic Interests: Artificial Intelligence | Energy Modeling | Uncertainty Modeling | Climate Applications | Low-resource Language

Personal Skills: Teamwork and Collaboration, Time management, Creativity, Adaptability, Excellent Communication Skills, Leadership

Languages: English, Yoruba

PROFESSIONAL MEMBERSHIPS

- **Graduate Member**, Nigeria Society of Engineers (NSE)
- **Member**, Data Science Nigeria (DSN)

VOLUNTEERING ACTIVITIES

2023 - Present	Maintainer & Documentation Lead, PyPSA-Earth Community • As a maintainer, I ensure the stability of the PyPSA-Earth framework by reviewing community contributions while simultaneously leading a strategic initiative to overhaul the project's documentation for improved user accessibility.
2024 - Present	Data Engineer Mentor, Data Epic • Mentored aspiring data engineers by providing expert guidance on industry best practices and assisting with hands-on projects to build career-ready, practical skills.
2022 - 2023	Cowrywise Student Ambassador, Cowrywise, Lagos, Nigeria • Championed financial literacy among university students while acting as a community liaison to inform the development of youth-focused financial products.
2021	Data Science Instructor, DSN Campus Invasion Program, Nigeria • Instructed university students on the fundamentals of data science, from analysis to introductory machine learning, through interactive sessions and hands-on projects.
2016 - 2024	Chess Instructor, HydeasHub, Lagos, Nigeria • Developed comprehensive chess programs and organized tournaments for kids in my local community.

PROJECTS

2025	Curation of Yoruba-English Code-Switching Speech Data Curated a novel, high-quality Yoruba-English code-switched speech corpus through systematic collection, segmentation, and transcription to enable robust research in low-resource ASR systems.
2025	PyPSA Helper Bot (Discord Bot) Developed a specialized Discord bot to provide automated support and streamline information access for the global PyPSA open-source energy modeling community.
2024	Solsight – Serverless File Setup Contributed to an enterprise-grade serverless data pipeline by architecting and implementing robust logic for data ingestion, cleaning, and transformation to ensure workflow reliability at scale.
2023	GitHub Archive Data Engineering Pipeline Engineered a scalable, end-to-end batch data pipeline on GCP using Terraform for automated provisioning and PySpark/dbt for transformation to deliver key analytics on GitHub activity via Looker Studio.
2023	Nigerian Housing Price Prediction Software Architected a full-stack MLOps pipeline for housing price prediction, integrating MLflow for model versioning and a CI/CD workflow with Prefect and Docker for automated deployment and real-time monitoring.
2023	Podcast Summarizer Software Developed and deployed a podcast summarization tool by fine-tuning a specialized language model and integrating it into an interactive Streamlit frontend using Modal Labs for serverless hosting.